MIL-I-5417D
13 November 1968
Superseding
MIL-I-5417C
4 October 1965

MILITARY SPECIFICATION

INDICATOR, INDICATED AIRSPEED, 40-400 KNOTS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of pitot static airspeed indicator, range 40 to 400 knots designated MS28046.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawing, Engineering and Associated List
MIL-I-5415	Indicators; Airspeed, Pitot and Static Pressure, General Specification for
MIL-N-7513	Nomenclature Assignments, Contractors Methods for Obtaining
MIL-L-25142	Luminescent Material, Fluorescent

STANDARDS

Federal

FED-STD-595	Colors

FSC 6610

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Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-831	Test Reports, Preparation of
MS28046	Indicator, Indicated Airspeed, Pitot Static 40-400 Knots
MS33585	Pointer, Dial, Standard Design of Aircraft Instruments

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

* 3.1 First article. This specification provides for first article inspections.

3.2 General. The general requirements specified in MIL-I-5415 are applicable as general requirements of this specification. Where the requirements of MIL-I-5415 and this specification conflict, the requirements of this specification shall govern.

3.3 Material. Material shall be in accordance with the requirements specified in MIL-I-5415.

3.4 Design and construction.

3.4.1 Range. The airspeed indicator shall be designed to satisfactorily operate in the range of 40 to 400 knots.

3.4.2 Dimensions. The airspeed indicator shall conform to the dimensions shown on MS28046.

3.4.2.1 Case finish. The case shall be finished with a lusterless black material, color 37038 of FED-STD-595. The finishing material shall be of a durable type to withstand usage encountered in service.

3.4.3 Pointer. The pointer shall conform to MS33585-3. The shaded portion of the pointer shall be finished in fluorescent material conforming to MIL-L-25142 or lusterless white, color 37875 of FED-STD-595. The unshaded portion of the pointer shall be finished in lusterless black, color 37038 of FED-STD-595.

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3.4.3.1 Limitation of pointer movement. The pointer movement shall be limited by suitable stops in the mechanism in such a way that the pointer will not be permitted to rotate more than 355° of arc clockwise from the zero graduation, when the indicator is submitted to overpressure tests. Suitable stops may also be incorporated in the indicator mechanism to limit the counter clockwise rotation of the pointer to the zero graduation. The design of the stops shall be subject to the approval of the procuring activity.

3.5 Weight. The weight of each indicator shall not exceed 1 pound.

3.6 Performance. The airspeed indicators shall satisfactorily meet the requirements specified in MIL-I-5415.

* 3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-1000 shall govern the manufacturer's part numbers and changes thereto.

3.8 Markings.

3.8.1 Dial. Dial marking shall be durable, and the size, material, and finish shall conform to Figure 1.

3.8.2 Fluorescent luminescent material. The fluorescent luminescent material shall conform to MIL-L-25142.

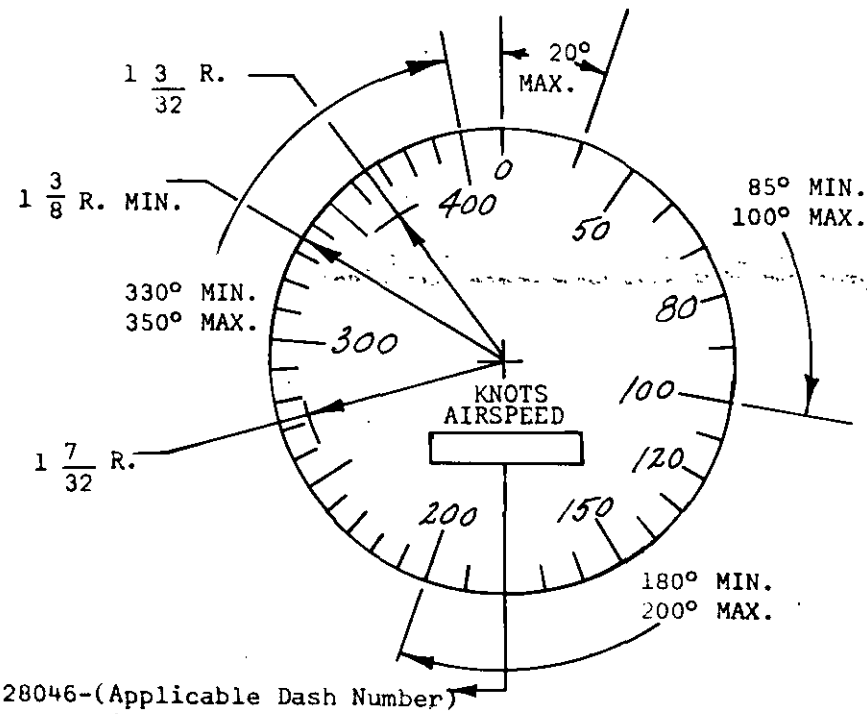
3.8.3 Lusterless material. The lusterless white material shall be color 37875 of FED-STD-595. The lusterless black material shall be color 37038 of FED-STD-595.

3.8.4 Visibility of dial markings. All graduations, numerals, pointers, and other markings on the dial shall be readable from any point within the frustum of a cone, the sides of which makes an angle of not less than 30° with a perpendicular to the dial and the small diameter of which is the aperture of the dial.

3.9 Identification of product.

3.9.1 A nameplate conforming to MIL-N-7513 shall be securely attached to the exterior of the case and shall be marked in accordance with the requirements of MIL-STD-130.

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MS28046-(Applicable Dash Number)
All arc dimensions shall start at the zero position.
All graduations are partial radial lines.
Dimensions in inches. Unless otherwise specified, tolerances: Fractions ± 1/64

MARKING	HEIGHT OR LENGTH ± 0.010	WIDTH OF LINE OR GRADUATION ± 0.005	MATERIAL OR FINISH
Numerals 50, 80, 100, 120, 150, 200, 300, 400	0.188	0.031	Fluorescent Luminescent Material or Lusterless White
Graduations corresponding to numerals 50, 100, 150, 200, 250, 300, 350, 400	As shown on figure	0.031	
Graduations other than those specified	As shown on figure	0.016	
Numeral "0"	0.156	0.016	
Graduation corresponding to "0"	0.125	0.016	
Lettering "Knots"	0.188	0.016	
Lettering "MS28046"	0.047	0.016	Lusterless Black
Lettering "Airspeed"	0.047	0.016	
Background of Dial	- -	- -	

FIGURE 1. Dial markings

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TABLE I
Test Values

Tests	Tolerances (Values in knots unless otherwise specified)
DAMPING	
Time required from full scale deflection to 50 knots	1.0 $^{-0.2}_{+0.5}$ second
LOW TEMPERATURE EXPOSURE AND OPERATION	3.5
HIGH TEMPERATURE OPERATION	2.5
AGING	2.5
VIBRATION FAILURE	
Test differential pressure equivalent to 170 knots	2.5
FRICTION	
Between 40 and 200 knots	3.5
Between 201 and 400 knots	5.0
POSITION ERROR	2.5
POINTER ZERO POSITION	
Indication at 0	1/16 inch
VIBRATION ERROR	
Test differential pressure equivalent to 120 and 240 knots	2.5
OVERPRESSURE	
Pitot pressure connection - 140 inches of water	2.5
Static pressure connection - 6.0 inches of water	2.5
SEASONING	2.5

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TABLE II
Scale Errors At Room Temperature

Speed (Knots)	Pressure (Inches of Water)	Allowable Scale Errors (Knots)
40	1.045	2.5
50*	1.634	2.0
60*	2.354	2.0
70	3.207	2.0
80	4.192	2.0
90	5.310	2.0
100*	6.563	2.0
110	7.951	2.0
120	9.475	2.0
130	11.14	2.0
140*	12.94	2.5
150	14.87	2.5
160*	16.95	2.5
170	19.17	3.0
180	21.54	3.0
190	24.05	3.5
200*	26.71	4.0
210	29.51	4.0
220	32.47	4.0
230	35.58	4.0
240	38.84	4.0
250*	42.27	4.0
260	45.85	4.0
270	49.59	4.0
280	53.50	4.0
290	57.58	4.0
300*	61.82	4.0
310	66.24	5.0
320	70.83	5.0
330	75.61	5.0
340	80.56	5.0
350*	85.70	5.0
360	91.04	5.0
370	96.56	5.0
380	102.28	5.0
390	108.20	5.0
400*	114.33	5.0

* Test points.

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3.10 Workmanship. Workmanship shall be in accordance with MIL-I-5415.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (See 4.3)

b. Quality conformance inspection (See 4.4)

* 4.3 First article inspection.

* 4.3.1 Test sample. The test sample shall consist of two indicators representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity. The samples shall be prepared as follows:

* 4.3.2 Data to accompany test sample. The test sample shall be accomplished by the following data:

a. Brief operating data to enable test personnel to correctly operate the equipment.

b. Engineering data in the form of assembly drawings (2 sets) and calibration test of the indicator.

* 4.3.3 Test report, test sample and data for the procuring activity. When the tests are conducted at a location other than the laboratory of the procuring activity, the following shall be furnished to that activity:

a. Test report. Three copies of a test report in accordance with MIL-STD-831.

b. Test sample. The sample which was tested.

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c. Data. The data required under paragraph 4.3.2.

* 4.3.4 First article inspection. The first article inspections shall consist of all tests described under "test methods", paragraph 4.6 of specification MIL-I-5415.

* 4.4 Quality conformance inspections.

* 4.4.1 Samples of the indicators shall be selected in accordance with MIL-I-5415. The inspection and testing of indicators shall be in accordance with MIL-I-5415, except that the test points, allowable error, and pressure calibration values shall be as shown in Tables I and II of this specification.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging.

5.1.1 Level A. Indicators shall be packaged in accordance with Method 1A of MIL-P-116 one each in a container conforming to PPP-B-636, weather-resistant class.

5.1.2 Level C. Indicators shall be provided sufficient protection against physical and mechanical damage to insure safe delivery, without degradation of the item reliability, from the supply source to the first domestic receiving activity for immediate use.

5.1.3 Levels of packaging and packing. The levels of packaging and packing shall be as specified by the procuring activity. See paragraph 6.3.

5.2 Packing.

5.2.1 Level A. Indicators packaged as specified in 5.1.1 shall be packed in fiberboard boxes conforming to PPP-B-636, weather resistant class, VIS grade for shipments not exceeding 70 pounds. Shipments exceeding 70 pounds shall be packed in cleated-plywood boxes conforming to PPP-B-601, overseas type, surface treated in accordance with the requirements of the specification.

5.2.2 Level B. Indicators packaged as specified in 5.1.1 shall be packed in domestic class exterior containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 shall apply.

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5.2.3 Level C. Packing which affords adequate protection against damage during direct domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the suppliers commercial practice, provided the later meets the requirements of this level.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Precautionary marking. The following precautionary marking shall appear on two opposite sides of each interior package and shipping container whenever practicable depending on the size of the carton:

FRAGILE

DELICATE INSTRUMENT

HANDLE WITH CARE

6. NOTES

6.1 Intended use. The intended use contained in section 6 of MIL-I-5415 is applicable to and forms a part of this specification.

6.2 General. The instructions contained in section 6 excluding provisions for qualification tests of MIL-I-5415 are applicable to and form a part of this specification except as specified herein.

6.3 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. The quantity, MS part number and class number of the instrument desired.

c. Dial markings and boss desired.

d. Levels of packaging (5.1) and packing (5.2) desired.

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* NOTE: " The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue."

Custodians:
Air Force - 71
Navy - AS
Army - AV

Reviewer:
Army - AV
Navy - AS
Air Force - 71

Preparing activity:
Air Force - 71

Project No. 6610-0295

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE Air Force OC, 29 Apr 64-400

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